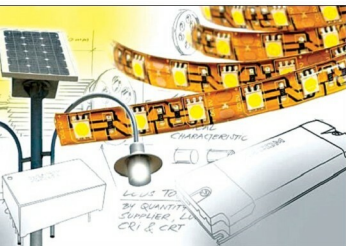


# LED Drivers: The Engines Behind New Lighting Systems



Make no mistake—LEDs are the lighting devices of the future! With LEDs, lighting becomes electronic—replacing a technology that has seen few changes over the past 100 years or so. Light-emitting diodes are powered by low direct current and, therefore, do not easily fit into our mains systems. To operate them, you need efficient drivers that convert the power from the wall socket to the power level required for LEDs. As most drivers are not visible, their role is rarely recognised. However, for the lighting systems of the future, they are as important as engines are for cars. Nothing works without them

## STEPHEN WEGSTEIN

**I**ncandescent lamps are on the way out and are being replaced by energy-saving CFL bulbs. Only a few years ago, these devices were hailed as the solution to our energy problems. Since then, they have lost some of their sheen as it emerged that dangerous mercury vapour gets released when the glass is broken. In addition, it takes a while for them to warm up and to emit their full light. Besides, the energy-saving bulbs cannot be dimmed.

The McKinsey report on energy efficiency is probably right in predicting that LED technology will replace energy-saving bulbs in a few years. There has been tremendous progress in the development of LED technology over the past five years.

Today, electronic lighting systems are leading the field when it comes to efficiency and lifetime. In addition, they offer designers virtually unlimited new possibilities in lighting control and luminaire design. Just as in cars, where the body around the engine can be designed in any shape or colour, LED drivers play a crucial role in lighting up the world around us. In a car, the energy contained

in the fuel is converted by the engine into movement. LED drivers convert any given voltage into a constant current that makes the LEDs light up in the desired brightness and colour.

## LED technology at a glance

Each LED requires a threshold voltage of just above 3V. As this value varies slightly even between LEDs with the same design, the luminous efficacy is always proportional to the actual current. LEDs are generally installed in series and powered by a constant current. All LEDs are thus lit with the same brightness, which is important for homogenous lighting.

As LEDs are available with different power ratings, the drivers must produce different constant currents to operate them. One-watt LEDs require 350mA, while 2W LEDs need 700mA. The maximum output voltage of a driver thus determines the number of LEDs that can be operated in series.

There are currently various products known as 'multichip arrays' that combine a number of low-power 'LED dies' on a 2-square-centimetre ceramic disk. The disk is covered in a phosphorus coating so that the individual

light sources appear as one. Multichip arrays require a voltage of around 40V and a constant current of 500 or 700mA, depending on the power rating.

Another type of LED lighting device is the flexible light strip, which is available in any length and is often used for indirect lighting. Three or six LEDs are combined into short strings that are supplied with constant current through tiny driver chips. Several such strings can be combined to form light strips. To supply these light strips, special lighting-class power supplies that are certified according to EN 61000-3-2 class C are required.

Apart from the standard 230V AC power from the electricity mains, many applications with drivers are powered by direct current. When the voltage is high enough to supply the desired number of LEDs, buck drivers can be used. When the voltage is too low, boost drivers are required. However, in some cases, a combined driver type is needed.

## Greater flexibility thanks to new buck/boost drivers

In battery-operated lighting systems, there is often the problem of the voltage dropping as the battery becomes discharged. Let us assume that we wish to